

Taxation and the Welfare State: Solutions to Problem Set 11

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Question 1)

Let's consider the proposed measures one-by-one:

i.

We know from empirical evidence that increasing the NRA has a large effect on retirement behavior (Seibold, 2021). Hence, this measure will be effective at encouraging later retirement, which improves the fiscal balance of the pension system.

ii.

We know from empirical evidence that increasing the ERA also has a large effect on retirement behavior (Seibold, 2021). Hence, this measure is going to be effective at encouraging later retirement and improving the fiscal balance of the pension system. Note that the main difference between i. and ii. is whether early or late retirees are targeted by the reform.

iii.

We know from empirical evidence that marginal financial incentives have only a modest effect on retirement behavior (Manoli & Weber 2016; Seibold 2021). Increasing statutory retirement ages (i. and ii.) will likely be a more effective policy than increasing marginal financial incentives.

iv.

Reducing future pension levels (or letting them grow less fast) improves the fiscal balance both because it mechanically reduces expenditure and because it induces later retirement (Gelber et al. 2017). This measure will thus also contribute to fiscal consolidation.

v.

Population aging is a particular problem for pay-as-you-go pension systems. A fully funded system is more robust because its returns are decoupled from population growth. Thus, introducing funded components will help increase future pension levels. But current workers have to pay higher contributions, as they are paying for both the pay-as-you-go and the funded component at the same time.

Question 2)

a.

$$\max_{c_y, c_o} \sqrt{c_y} + \sqrt{c_o} \quad s.t. \quad c_y = 2$$

When people cannot save privately, the budget constraint is simply $c_y = 2$, which pins down the solution $c_y = 2$, $c_o = 0$. The rationale for a public pension program in this case would be a lack of private savings opportunities.

b.

$$\begin{aligned} c_y^1 &= (1 - t) \cdot 2 = 50\% \cdot 2 = 1 \\ c_o^0 &= b^0 = t \cdot 2 \cdot \frac{N^1}{N^0} = 50\% \cdot 2 \cdot 1.02 = 1.02 \end{aligned}$$

In the first line above, we calculate consumption of generation 1 at young age c_y^1 . In the second line, we calculate the pension benefit paid to old individuals, which corresponds to consumption of generation 0 at old age c_o^0 . Here we need to plug in the population growth rate $1 + n = N^{t+1}/N^t$.

Next, we can calculate the utility of individuals without and with the public pension program. Without the pension program, utility is

$$u^t = \sqrt{2} + \sqrt{0} = \sqrt{2}$$

After the program is phased in, utility is

$$u^t = \sqrt{1} + \sqrt{1.02} > \sqrt{2}$$

Hence, people are better off as the pension program allows them to smoothe consumption.

Given the timeline of the introduction of the program, we can calculate each generation's consumption:

$$\begin{aligned} c_y^0 &= 2 \\ c_o^0 &= 1.02 \\ c_y^1 &= 1 = c_y^2 = \dots \\ c_o^1 &= 1.02 = c_o^2 = \dots \end{aligned}$$

Generations 1, 2, etc. gain utility as shown above. Generation 0 experiences the greatest gain, as they did not pay contributions when young but received benefits when old. This is at the source of the “legacy debt” of the PAYG pension system.

c.

Let’s find the tax/contribution rate that maximizes individuals’ utility under $r = 5\%$.

$$\max_{c_y, c_o} \sqrt{c_y} + \sqrt{c_o} \quad s.t. \quad c_y + s = 2, \quad c_o = 1.05 \cdot s$$

The problem can be re-written as

$$\max_s \sqrt{2 - s} + \sqrt{1.05s}$$

The first-order condition is

$$-\frac{1}{2\sqrt{2-s}} + \frac{\sqrt{1.05}}{2\sqrt{s}} = 0$$

Re-arranging yields $s = 2.1/2.05$. The contribution rate is defined relative to income at young age: $t = s/2 \approx 51.2\%$.

Question 3)

a.

True. A PAYG pension system itself does not involve savings, as contributions of the young are directly transferred to the old. A funded system, on the other hand, mandates individuals to save for retirement, building a stock of aggregate savings. Thus, when switching from PAYG to funded, aggregate savings should increase. By how much savings increase depends on the crowd-out of existing private savings that people may have had already under the PAYG system.

b.

In principle true. If the PAYG system is terminated at some point, legacy debt is a problem – the last generation who contributed is “owed” a pension. One way to deal with this problem would be for the government to increase borrowing in order to finance pension payments to this generation, thus distributing the debt payments across multiple future generations. However, if the PAYG system continues indefinitely, the legacy debt could be carried through forever, such that it may not present a problem.

c.

Ambiguous. PAYG pension schemes yield high returns when the rate of real wage growth and/or the rate of population growth are high. Viewed from this angle, they may appear as a good choice for developing countries. However, PAYG systems require political stability and

trust to work. Current workers pay contributions and receive a promise of future pensions, but in principle the pension system could be changed at any time. This differs from a funded system, where each individual is entitled to the stock of savings they accumulated.

d.

False. When there are differences in life expectancy across groups, pension schemes paying equal benefits to different groups of people do not ensure that everyone gets the same net present value of pensions. For instance, women have higher life expectancy than men on average and thus receive a higher NPV for given monthly pension benefits.

e.

In principle true. According to the standard life cycle model, people desire to smooth consumption over time. Under the optimal consumption path, there should be not be a drop in consumption at any age. Hence, an observed drop in consumption at retirement indicates that individuals did not save enough for retirement. However, some critics have argued that we have to be careful with this interpretation. There may be desired drops in expenditure at retirement, for instance due to increased home production, reduced work-related expenditures, etc.

f.

False. There is a moral hazard cost associated with PAYG pension systems. When pension benefits are available at different ages, some individuals may retire “too early”, taking advantage of early retirement benefits intended to provide insurance against low earnings ability at old age. An optimal PAYG pension system has to trade off the gains from insurance vs. the moral hazard cost of early retirement.